



Technical Work Breakdown & System Rules

Deadline 20/6

1 System Architecture Overview

The system is designed as a modular, layered architecture to ensure clarity, scalability, and strict control over responsibilities. It consists of **five primary layers**, each with a clearly defined role and set of rules:

1. Mobile Application Layer
2. Backend API & Business Logic Layer
3. AI & Machine Learning Layer
4. Generative AI Communication Layer
5. Data Storage & Security Layer

This separation ensures maintainability, ethical control, and predictable system behavior.

2 Mobile Application Layer

2.1 Responsibilities

The mobile application serves as the user-facing interface and is responsible for:

- Handling all user interactions
 - Collecting user inputs, including questionnaire responses
 - Presenting mood analysis results and self-care guidance
 - Visually enforcing recovery window restrictions
 - Displaying historical mood insights in a clear and calm manner
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2.2 Technologies

- Flutter (Dart) or React Native for cross-platform development

- Minimalistic and calming UI design principles
 - Local state management to reflect recovery window status
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2.3 Core User Interface Screens

- Welcome & Disclaimer Screen
 - Home Screen (“Check how I feel”)
 - Mood Assessment Questionnaire
 - Mood Result & Guidance Screen
 - Recovery Window Timer Screen
 - Mood History & Summary Dashboard (basic)
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Mobile UI Rules

1. Medical terminology such as “diagnosis” or “treatment” must not appear.
 2. All language must be supportive and neutral (e.g., “guidance”, “suggestion”).
 3. The questionnaire must be inaccessible during an active recovery window.
 4. The user must always clearly understand:
 - Their current mood state
 - The recommended actions
 - The estimated recovery duration
 5. Notifications must never increase anxiety or urgency.
 6. Visual design must prioritize calm colors, minimal text, and cognitive simplicity.
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Backend System (API & Business Logic)

3.1 Responsibilities

The backend acts as the **central decision authority** of the system. Its responsibilities include:

- Receiving and validating questionnaire responses
- Orchestrating AI model execution
- Applying business rules and ethical constraints
- Persisting user data securely
- Controlling reassessment permissions and recovery windows

3.2 Technologies

- Python-based backend using FastAPI or Flask
 - RESTful API architecture
 - Basic authentication or anonymous session handling (MVP)
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3.3 Core Backend Modules

1. User Management Module

- Anonymous or basic user identification
- User preferences
- Mood assessment history

2. Assessment Processing Module

- Receives questionnaire inputs
- Normalizes and validates responses
- Converts inputs into ML-ready features

3. Decision Engine

- Applies business logic for:
 - Mood classification
 - Severity scoring
 - Recovery window calculation

4. Recovery Control Module

- Prevents reassessment during recovery windows
 - Allows reassessment only through explicit user override
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Backend Rules

1. The backend is the single source of truth for system decisions.
2. The frontend must not override recovery window restrictions.
3. All system decisions must be explainable and traceable.
4. No real-time or continuous monitoring is permitted.
5. Every AI-generated output must pass rule-based validation before delivery.

4 AI & Machine Learning Layer

4.1 Purpose

This layer provides analytical intelligence by estimating:

- The user's current mood state
 - The severity of the mood
 - Repeated or persistent negative patterns over time
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4.2 AI Components

1. Mood Classification Model

- **Input:** Questionnaire responses
- **Output:** Mood category (e.g., stable, stressed, anxious)

MVP Models:

- Logistic Regression
- Random Forest

Rationale:

- High interpretability
 - Suitable for small datasets
 - Low computational overhead
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2. Severity Scoring Mechanism

- Combination of rule-based logic and statistical scoring
- Based on:
 - Weighted questionnaire answers
 - Response frequency
 - Recent assessment history

Output Levels:

- Low
- Medium

- High
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3. Lightweight Pattern Detection

- Analyzes the last 3–7 assessments
 - Detects repetition of negative mood states without diagnosis
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AI Rules

1. No clinical or diagnostic labels are permitted.
 2. The system must not predict mental disorders.
 3. AI outputs must represent current-state estimation only.
 4. Thresholds must remain conservative by design.
 5. AI models must never operate autonomously.
 6. Rule-based logic must always constrain AI outputs.
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Recovery Window Logic (Core Innovation)

5.1 Recovery Time Calculation

Recovery duration is calculated using:

- Identified mood type
- Severity level
- Number of recommended actions
- Estimated duration of each action

Example:

Mood: Stress (Medium)

Breathing Exercise: 5 minutes

Relaxation Technique: 8 minutes

Short Break: 10 minutes

Estimated Recovery Window: 25–35 minutes

5.2 System Behavior During Recovery

- Mood reassessment is disabled
- A countdown timer is displayed

- Users are encouraged to focus on recommended actions
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Recovery Window Rules

1. Reassessment is blocked until the recovery window ends.
 2. Early reassessment is allowed only if the user reports worsening feelings and is shown a safety disclaimer.
 3. Recovery time estimates are informational, not guarantees.
 4. Recovery windows are supportive guidance, not enforced restrictions.
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Generative AI Communication Layer

6.1 Role

Generative AI is used strictly for communication purposes, including:

- Explaining mood analysis results
 - Describing self-care actions
 - Providing emotionally supportive and reassuring language
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6.2 Explicit Limitations

Generative AI must not:

- Provide diagnoses
 - Suggest treatments or medications
 - Handle emergency or crisis scenarios
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6.3 Prompt Governance

Prompts must be:

- Structured and constrained
- Safety-aware
- Reviewed and validated by backend rules

Example Instruction:

“Explain the result in calm, supportive language. Avoid medical advice. Encourage self-care and professional support when appropriate.”

Generative AI Rules

1. All generated text must pass backend validation.
 2. Absolute claims (e.g., “you will feel better”) are prohibited.
 3. Probabilistic language must be used (“often helps”, “may support”).
 4. Dependency-inducing language is not allowed.
 5. Crisis-related content must redirect users to external resources.
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7 Data Storage & Security Layer

7.1 Stored Data

- Questionnaire responses
 - Mood classifications and severity scores
 - Timestamps
 - Optional user feedback
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7.2 Data Explicitly Excluded

- Medical records
 - Audio or video data
 - Passive sensor data
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Data Governance Rules

1. Apply strict data minimization principles.
 2. Encrypt all stored data at rest.
 3. Use anonymous identifiers for MVP.
 4. Users must be able to delete their data.
 5. No data sharing with third parties.
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8 Ethical & Safety Framework

Ethical Foundations

- Clear disclaimer on first application launch
 - No replacement of professional mental health services
 - No claims of emergency handling capability
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Ethical Rules

1. The application functions as a companion, not a therapist.
 2. AI assists users but never makes decisions for them.
 3. User autonomy and control are preserved at all times.
 4. Professional help is encouraged and never delayed.
 5. No addictive engagement or compulsive usage design.
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Development Workflow

Project Phases

1. UX design and questionnaire formulation
 2. Backend development and AI logic implementation
 3. Mobile application integration
 4. System testing and edge-case validation
 5. Final demonstration and evaluation
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Final Summary

This project is implemented as a modular AI-powered mental health companion where the mobile application manages user interaction, the backend enforces functional and ethical constraints, machine learning models estimate mood and severity, and generative AI delivers supportive communication. A recovery window mechanism ensures realistic emotional processing by preventing repeated assessments. The entire system is designed with strict safety, privacy, and ethical considerations, enabling meaningful mental health support without medical claims.
